

NGUYEN TAN

Senior Backend Engineer

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Professional Summary

Backend engineer with **5+ years** of experience building **high-performance, distributed systems** using **Golang** and cloud-native technologies. Proven track record in designing **event-driven microservices architectures** handling millions of concurrent users, with deep expertise in **GCP, Kubernetes, gRPC, and observability**. Experienced in fintech/banking domain with a strong focus on **system reliability, latency optimization, and CI/CD automation**. Passionate about open-source and building resilient systems at scale.

Technical Skills

Languages:	Golang (primary), TypeScript/JavaScript, Python, SQL
Backend & APIs:	gRPC, REST, WebSocket, GraphQL, Protocol Buffers
Databases:	PostgreSQL, MongoDB, ScyllaDB/Cassandra, Redis, MySQL
Message Brokers:	Apache Kafka, RabbitMQ, Google Pub/Sub
Cloud & Infra:	GCP (GKE, Cloud Run, Cloud SQL, Pub/Sub, BigQuery), Docker, Kubernetes, Istio
Observability:	Prometheus, Grafana, OpenTelemetry, Jaeger, structured logging (Zap/Logrus)
CI/CD & DevOps:	GitHub Actions, ArgoCD, Terraform, Helm, Jenkins
Architecture:	Microservices, Event-Driven Architecture (EDA), Domain-Driven Design (DDD), CQRS
Testing:	Unit testing (Go testing, testify), integration tests, load testing (k6, wrk)
Other:	System Design, Chaos Engineering, Auto-scaling, Circuit Breaker patterns

Work Experience

Senior Software Engineer — Terralogic 2022 – Present

Backend-focused Full Stack Developer — Golang, GCP, Kubernetes, Kafka

- Architected and developed a **real-time chat platform** using **Golang microservices** with **gRPC** inter-service communication, supporting **3M+ concurrent WebSocket connections** with sub-100ms message delivery latency.
- Designed **event-driven architecture** using **Apache Kafka** for high-throughput message brokering, processing **500K+ events/second** across authentication, messaging, and notification services.
- Implemented **ScyllaDB** as the primary message store for time-series chat data, achieving **10x read performance** improvement over previous MongoDB-based solution.
- Integrated **Redis** cluster for session management, token caching, and user presence tracking with **99.9% cache hit rate**, reducing database load by **60%**.
- Deployed and operated services on **GKE (Google Kubernetes Engine)** with **Istio service mesh**, implementing auto-scaling policies, circuit breakers, and retry mechanisms for high availability.
- Built comprehensive **end-to-end observability** stack with **Prometheus + Grafana** dashboards, **OpenTelemetry** distributed tracing, and structured logging, reducing MTTR by **70%**.
- Designed and implemented **CI/CD pipelines** with GitHub Actions and ArgoCD for automated testing, container building, and progressive rollout with canary deployments.
- Developed a **Jenkins orchestration platform** (Node.js/Go) integrating real-time dashboards for CI/CD job monitoring, device management, and build analytics across **200+ daily builds**.
- Led **network device management dashboard** development — migrated legacy React codebase to modern architecture, integrated REST APIs for real-time device monitoring and control actions.
- Received **Snap Award 2024** for outstanding cross-functional contributions and technical leadership.

Full Stack Developer — Allgrow Labo

2021 – 2022

Web application development for Japanese market — React, Node.js, REST APIs

- Built backend services for **life insurance transaction platform**, implementing calculation engines for policy analytics and statistical reporting with **PostgreSQL** and optimized query patterns.
- Developed insurance campaign management modules with **RESTful API design**, handling complex business logic for policy pricing, underwriting rules, and transaction reconciliation.
- Implemented robust **unit and integration testing** strategy with Jest and React Testing Library, achieving **85%+ code coverage** and significantly reducing production incidents.

Frontend Developer (Intern) — *Xtek*

2021

Web development — React, CSS Architecture

- Built reusable component library with scalable CSS architecture, enabling team-wide code reuse and consistent UI patterns across multiple client projects.

Notable Projects & Open Source

High-Performance API Gateway (Personal Project)

Golang, gRPC, Redis

- Built a lightweight API gateway in Go with **rate limiting** (token bucket), **JWT authentication**, request routing, and **circuit breaker** pattern. Benchmarked at **50K+ req/s** on a single node.

Distributed Task Queue

Golang, Kafka, PostgreSQL, Docker

- Designed a distributed task scheduling system with **exactly-once processing** guarantees, dead-letter queues, and retry policies. Deployed on Kubernetes with **horizontal pod autoscaling**.

Real-time Chat System (Production)

Golang, WebSocket, Kafka, ScyllaDB, gRPC

- End-to-end ownership of a production chat platform: **system design, implementation, deployment, monitoring**. Handles real-time messaging with presence, typing indicators, and message persistence at scale.

Education

Posts and Telecommunications Institute of Technology — *Ho Chi Minh City*

2016 – 2020

Bachelor of Engineering in Computer Engineering

Certifications & Continuous Learning

- **Google Cloud Professional Cloud Architect** (in progress)
- Deep study: Distributed Systems, Event-Driven Architecture, Domain-Driven Design
- Active contributor to Go open-source ecosystem and internal knowledge-sharing initiatives
- AI-assisted development: Proficient with **GitHub Copilot**, **Cursor**, **Claude Code** for accelerated engineering workflows